



Editorial

Special issue on workflows in Support of Large-Scale Science



This special issue is based on the best papers from the 13th Workshop on Workflows in Support of Large-Scale Science (WORKS) [1], held in conjunction with the 2018 International Conference for High Performance Computing, Networking, Storage and Analysis (SC) [2], on November 11, 2018 in Dallas, TX, USA. The workshop focused on the many facets of data-intensive workflow management systems, ranging from job execution to service management and the coordination of data, service and job dependencies. It attracted a large number of scientific papers that contributed to the state-of-the-art in workflows research. All the papers selected for this special issue have been substantially extended from their original versions and underwent a rigorous peer-review process. This special issue is composed of eight highly rated research papers that cover a broad range of issues in the scientific workflow lifecycle, which include: stream-based workflow data processing in cloud environments, energy-aware workflow scheduling, transparent data provenance tracking, workflow simulation, IoT workflows, workflow performance data capture and analysis, and parallel pattern-based workflow management.

Throughput optimization for Storm-based processing of stream data on clouds [3]. In this paper, Cao et al. present techniques for modeling the behavior of Apache Storm based streaming data processing workflows and improving their performance in clouds. The paper constructs rigorous cost models to analyze the throughput dynamics of Storm workflows and formulates a budget-constrained topology mapping problem to maximize Storm workflow throughput in clouds. The authors show that this problem is NP-complete and design a heuristic solution that takes into consideration not only the selection of virtual machine type but also the degree of parallelism for each task (spout/bolt) in the topology. The performance advantages of the proposed mapping solution is illustrated through extensive simulations and further verified by real-life workflow experiments deployed in public clouds and compared with the default Storm and other existing methods.

Energy-aware workflow scheduling and optimization in clouds using bat algorithm [4]. In this paper, Gu et al. study the challenges faced in cloud environments including energy consumption, execution time, heat and CO₂ emission, as well as operational cost. Due to the extremely large scale of the data-intensive applications deployed on clouds, and a huge amount of resource consumption, even a small portion of the improvements in any of the above areas can yield huge ecological and financial rewards. The authors postulate that efficient and effective workflow scheduling in cloud environments is one of the most significant

ways to confront the above problems and to achieve optimal resource utilization. The authors propose an Energy Aware, Time, and Throughput Optimization heuristic (EATTO) based on the bat algorithm. The goal is to minimize energy consumption and execution time of computation-intensive workflows while maximizing throughput, without imposing any significant loss on the Quality of Service (QoS) guarantee.

Flux: Overcoming scheduling challenges for exascale workflows [5]. In this paper, Ahn et al. present a novel, hierarchical resource and job management software (RJMS) infrastructure, named Flux, addressing the key scheduling challenges of modern workflows in a scalable, easy-to-use, and portable manner. Flux is based on the principle that it can be seamlessly nested within batch allocations created by itself or other schedulers. Within each resource allocation, a hierarchy of Flux instances can be created within each allocation. The consistent and rich set of well-defined APIs in Flux can portably and efficiently support workflows that feature non-traditional execution patterns such as requirements for complex co-scheduling, massive ensembles of small jobs and coordination among jobs in an ensemble. The authors performed an extensive evaluation involving some of the emerging workflow efforts at Lawrence Livermore National Laboratory. The authors show that the proposed approach significantly addressing major workflow scheduling challenges such as, but not limited to, job throughput, co-scheduling, job coordination and communication and portability challenges. Further, their evaluation with synthetic and real-world ensemble-based workflows shows that their approach leads to improved job throughput performance.

Capturing and Analyzing Provenance from Spark-based Scientific Workflows with SAMbA-RaP [6]. In this paper, Guedes et al. present a solution for capturing, storing and querying prospective and retrospective provenance, and domain data within distributed scientific workflows named “Spark provenance Management with Reports and Presentation” (SAMbA-RaP). The proposed solution is motivated by the need to fill the gap between the Apache Spark in-memory Map-Reduce paradigm and the lack of support for both data tracking and workflow provenance. Before SAMbA-RaP, the users had to rely on spending countless hours collecting provenance data from log files. The authors show a performance evaluation of their proposed solution with distinct workflows and domains: the produced results show the overhead for managing provenance data did not exceed 4% of the considered baseline acting in a transparent way even in legacy applications scenarios.

Developing accurate and scalable simulators of production workflow management systems with WRENCH [7]. In this paper, Casanova et al. present a workflow management system (WMS) simulation framework, named WRENCH, performing accurate and scalable simulations and providing an easy simulation software development. The authors motivate their interest in simulating workflow applications because most of the WMS research and development activities rely on empirical experiments, which are performed with regular (production) software stacks on actual hardware platforms. These experiments, however, are limited to hardware and software infrastructures at hand and can be labor- and/or time-intensive. As a result, relying solely on real-world experiments impedes WMS research and development. The WRENCH system addresses this challenge. WRENCH is built on the SimGrid framework in order to achieve high accuracy and scalable simulations. The authors present two case studies where WRENCH is applied to simulate Pegasus WMS and WorkQueue application execution framework producing both qualitative and quantitative comparisons with other workflow simulators.

Efficient orchestration of Node-RED IoT workflows using a Vector Symbolic Architecture [8]. In this paper, Simpkin et al. present how their Node-RED system can be executed in a decentralized fashion for Internet of Things (IoT) workflows. The authors show how Node-RED workflows can be migrated into a decentralized execution environment for operation on mobile ad-hoc networks, and demonstrate this by converting a Node-RED based traffic congestion detection workflow to operate in a decentralized environment. The approach uses a Vector Symbolic Architecture (VSA) to dynamically convert Node-Red applications into a compact semantic vector representation that encodes the service interfaces and the workflow in which they are embedded. By extending existing services interfaces, with a simple cognitive layer that can interpret and exchange the vectors, the authors show how the required services can be dynamically discovered and interconnected into the required workflow in a completely decentralized manner. The resulting system provides a convenient environment where the Node-RED front-end graphical composition tool can be used to orchestrate decentralized workflows. This paper further extends this work by introducing a new dynamic VSA vector compression scheme that compresses vectors for on-the-wire communication, thereby reducing communication bandwidth while maintaining the semantic information content. This algorithm utilizes the holographic properties of the symbolic vectors to perform compression taking into consideration the number of combined vectors along with similarity bounds that determine conflict with other encoded vectors used in the same context. The resulting savings make this approach extremely efficient for discovery in service based decentralized workflows.

End-to-end online performance data capture and analysis for scientific workflows [9]. In this paper, Papadimitriou et al. present a system for analyzing the characteristics of scientific workflow applications to better understand the impact on the underlying infrastructure. The authors describe the architecture, the integration with existing well-established tools and newly developed tools to collect online performance statistics of workflow executions. The proposed solutions manage various, heteroge-

neous sources and homogenize them in a Elasticsearch based distributed database correlating online workflow performance data with data from the underlying infrastructure. The collected data is presented using an online dashboard. The authors validated the proposed approach executing real-world workflows under normal and abnormal conditions. The authors demonstrated that the proposed approach is able to detect the anomalies when artificially injected paving the path for analysis using automated machine learning based methods.

An efficient pattern-based approach for workflow supporting large-scale science: The DagOnStar experience [10]. In this paper, Sanchez-Gallegos et al. present DagOnStar (Directed Acyclic Graph On Anything), a lightweight Python library implementing a workflow paradigm based on parallel patterns that can be executed on any combination of local machines, on-premise high performance computing clusters, containers, and cloud-based virtual infrastructures. DagOnStar is designed to minimize data movement to reduce the application storage footprint. The authors explored a case study based on a real-world application to illustrate the use of this novel workflow engine: a containerized weather data collection application deployed on multiple infrastructures. The paper presents an experimental comparison with other state-of-the-art workflow engines, and shows that DagOnStar can run workflows on multiple types of infrastructure with an improvement of 50.19% in run time when using a parallel pattern with eight task-level workers.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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